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Litter thickness, but not root biomass, explains the average and spatial structure of soil hydraulic conductivity in secondary forests and coffee agroecosystems in Veracruz, Mexico

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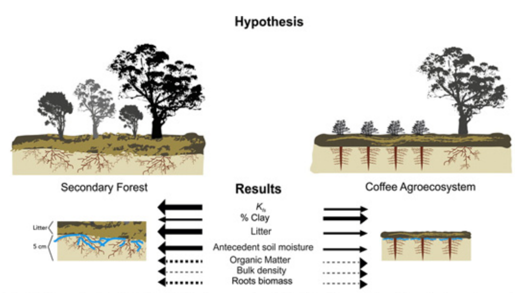
HIGHLIGHTS

- Saturated hydraulic conductivity (K_{fs}) was studied in secondary forest and coffee plots.
- Mean and spatial variation of litter and root biomass were analyzed with variography and linear models.
- K_{fs} occurred in a gradient at the landscape scale and, only in two sites, litter thickness was patchy within plots.
- Higher *Kfs* in secondary forests and its gradient in the landscape were explained by land use, litter thickness and clay.

GRAPHICAL ABSTRACT



Given that Secondary Forests (SF) and Coffee Agroecosystems (CA) differ in terms of vegetation structure and composition, we hypothesised that more abundant thick and medium roots in secondary forests would favour the formation of preferential flow channels, and a thicker litter would promote the conservation of soil moisture, explaining the higher capacity of the soil to conduct water (K_{fs}). Our results show that SF and CA differ in terms of several soil properties (thickness of the arrow indicates the magnitud of the difference), but only a few have explanatory power over K_{fs} (represented as solid arrows).

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ABSTRACT

Secondary forests and coffee agroecosystems are considered good alternatives for conservation of a high capacity for water filtration in the soil where tropical montane cloud forests (TMCF) once grew; however, it is not clear which characteristics of the vegetation modulate the field saturated hydraulic conductivity of the soil (K_{fs}) and whether these characteristics persist in such derived systems. Here, we explore how changes in vegetation between secondary forests and coffee agroecosystems have consequences for the average value and spatial variation of litter thickness and root biomass, and whether these differences can explain the K_{fs} and its spatial distribution. We found that the thickest litter, greatest total biomass and thickest roots are in the secondary forest of the north of the study area. The litter is spatially structured in patches of *ca.* 12 m at plot scale in the secondary forest and coffee agroecosystem of the southern area. Like the K_{fs}, the thickness of the litter and biomass of the thick (>2 mm), medium (1–2 mm) and fine (<1 mm) roots are spatially distributed on a north to south gradient at landscape scale. Our linear model indicates that geographic area (north or south), land use and litter thickness explain the K_{fs} and its spatial distribution along this gradient. Even on inclusion of the antecedent soil moisture and percentage of clays (found to explain K_{fs} in a previous study), it was not possible to eliminate from the model geographic area and land use, due to their high explanatory power. However, antecedent soil moisture became redundant on inclusion of the litter layer, which had a greater explanatory power. Our modeling suggests that undiscovered differences prevail between the geographic areas and secondary forest and coffee agroecosystems (possibly related to the edaphogenesis and management practices) that determine the K_{fs}.

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1. Introduction

Annual precipitation in tropical montane cloud forests (TMCF) can be between 500 and 10,000 mm (Hamilton et al., 1995). Of the total precipitation, intercepted mist can often account for 5 to 20% (Bruijnzeel and Proctor, 1995; Bugmann, 2001). With a high rate of infiltration and hydraulic conductivity (e.g., Marín-Castro, 2010 – 777.3 mm h^{-1}) in soils of volcanic origin, the TMCF conduct the water towards the rhizosphere to recharge the phreatic strata (Muñoz-Villers et al., 2015). Notwithstanding the recognized high water storage capacity of volcanic ash soils in TMCF, it remains unclear which vegetation characteristics modulate the entry and movement of water in the soil (Bruijnzeel, 2004; Roman et al., 2010). The high velocity at which water enters and is distributed within the initial layers of TMCF soils has been mainly attributed to their physico-chemical properties (Facelli and Pickett, 1991), such as silty clay textures, low bulk densities (<0.9 g cm^{-3}), high hydraulic conductivity (696 mm h^{-1}) (Muñoz-Villers et al., 2012), and high organic material content (4 to 15%) (Borman et al., 2007; Hillel, 2004; Hu et al., 2009; Sauer and Logsdon, 2002). However, various empirical studies have also highlighted the equal importance of the vegetation (Buytaert et al., 2005; Dörner et al., 2010; Zimmermann et al., 2006). For example, the infiltration and conductivity of water in the soil could be related to the thickness of the litter layer, which protects the soil from solar radiation, preventing moisture loss, laminar erosion and surface runoff (Gerke and Kuchenbuch, 2007; Van der Putten et al., 2013; Wu et al., 2016). Moreover, the high atmospheric humidity in the TMCF enables the litter to maintain the pores of the soil layers below in an active and moist condition such that, in events of precipitation and condensation of mist, the consequent infiltration of water through these pores is both rapid and deep (Cusack et al., 2009; Facelli and Pickett, 1991; Ilek et al., 2017; Walsh and Voigt, 1977).

Once past the litter layer, the flow of water from precipitation is absorbed by capillarity through the pores of the soil matrix or by preferential flow, enabled by the macroporosity of the channels formed by roots and edaphic fauna (Alaoui and Goetz, 2008; Meek et al., 1992). These channels increase the speed and control main direction of the flow during wetting and drying cycles (Skovdal Christiansen et al., 2004). In some cases, it has been reported that 74 to 100% of the water flow can occur through the channels of thick roots (Alaoui and Helbling, 2006). The roots also modify the physico-chemical properties of the soil, producing exudates that form organo-mineral complexes that facilitate soil respiration and decompaction (Pierret et al., 2007). Even the root structure itself exerts pressure on the pores, ultimately favoring slope stability (Ghestem et al., 2011; Stokes et al., 2009). Despite the recognized importance of the organic soil horizons to the hydric balance of these forests, little is known about how and to what extent anthropogenic transformation of the forests affects the hydraulic properties of their soils.

The severe fragmentation and transformation of TMCF in recent decades has been documented to lead to wide variation in their capacity for water capture and storage (Ataroff and Rada, 2000; Marín-Castro et al., 2016; Muñoz-Villers et al., 2012; Zimmermann and Elsenbeer, 2008). In Mexico and worldwide, large areas of forest have been transformed, mainly into pastures, silvicultural plantations, sugarcane monocultures, coffee plantations under different shade managements and secondary forests (Aide et al., 2010; Bubb et al., 2004; Toledo-Aceves et al., 2011; Williams-Linera et al., 2002). Of all of these derived systems, secondary forests with at least 20 years of natural regeneration have been recognized as bearing the greatest similarity to the original forests, in terms of the structure and diversity of the tree-shrub vegetation and capacity to capture and store water (Muñoz-Villers et al., 2015), although the time to recovery of the hydro-physical properties in secondary forests can vary according to region (Zimmermann et al., 2010). Efforts of conservation directed towards hydric resources have also highlighted the importance of other agroecosystems, such as coffee plantations (Ponette-González et al., 2010), due to the fact that these systems conserve different species of native trees for crop shading, as well as considering the capacity of their vegetal cover to intercept precipitation and reduce transpiration (Holwerda et al., 2013). Despite the similarities in tree-shrub structure between shade coffee plantations and secondary forests and between secondary forests and TMCF, it remains unknown whether the accumulation of litter and the structure of roots differ among transformed ecosystems and whether these characteristics are related to their capacity to capture and conduct water.

In shade coffee agroecosystems, the shrub stratum is normally dominated by a single species (*Coffea* sp.), a fact that could diminish the supply of organic material and the formation of litter (Kumar, 2007), relative to that of secondary forest, which is often more heterogeneous (Negrete-Yankelevich et al., 2006). As a result of this dominance of coffee plants, the coffee agroecosystems have a greater abundance of medium to fine (<1–2 mm) roots concentrated in the first 30 cm of soil (Defrenet et al., 2016; FAO, 2017): This is in contrast to forests, where the rhizosphere is often dominated by the thick roots of mature trees (Soethe et al., 2006) and a greater abundance of medium roots belonging to the shrubs.

It is believed that the entry of water into the soil under the influence of the vegetal coverage (litter and rhizosphere), or edaphogenetic changes, can be structured in different spatial scales (Litaor et al., 2002). For example, the influence of specific species on litter quality and of root structure could generate patch-like patterns at scales of tens of meters, a phenomenon known as home-field effect (Corti et al., 2002; Freschet et al., 2012; Negrete-Yankelevich et al., 2007; Titus and del Moral, 1998). At landscape scale (hundreds of meters), however, the mosaic of land uses, changes in soil type and topography can generate much larger patches and gradients.

In a previous study in central Veracruz, Mexico (Marín-Castro et al., 2016), we found that the capacity of the soil to conduct water (K_{fs}) is higher in secondary forests than in coffee agroecosystems and that K_{fs} presents a spatial distribution on a gradient at landscape scale but random within the plots. While we found that this difference in K_{fs} can be partly explained by differences in the antecedent moisture content and clay content of the soil, these variables alone are insufficient to explain all of the differences between these ecosystems. Given that the secondary forests and coffee agroecosystems differ in terms of vegetation structure and composition, we tested the following hypotheses: 1) the secondary forest will have a thicker litter layer and greater biomass of thick and medium roots than the coffee agroecosystems; in contrast, the coffee agroecosystems will have a greater biomass of fine roots, due to the fact that most of the roots will belong to the coffee plants; 2) where the thick and medium roots influence the formation of preferential flow channels, and the litter influences the conservation of soil moisture, both litter thickness and the biomass of thick and medium roots will explain the average variation in the hydraulic conductivity of these ecosystems; and 3) the spatial structuring of litter thickness and root biomass will be in agreement at landscape scale and that these variables will statistically explain the spatial variation of K_{fs}.

2. Material and methods

2.1. Study area

The study site is located in the lower part of the sub-basin of the Los Gavilanes River, between coordinates 19° 28′36″N, 97° 00′ 24″W and 19° 27′ 27″N 96° 58′ 58″W, in Veracruz, Mexico (Fig. 1a). Elevation ranges from 1200 to 1400 m. a. s. l., the climate is of type C (fm) temperate humid (García, 2004), with an annual mean temperature of 19 °C, annual mean precipitation of 1385 mm and annual evapotranspiration of 1120 mm (Muñoz-Villers et al., 2016). The climate is divided into dry (November–April) and rainy (May–October) seasons (Muñoz-Villers and McDonnell, 2012). For this reason, two sampling areas, known as north and south, were delimited in the lower part of the subbasin. These areas divided by the Los Gavilanes river and both correspond to low but steep slopes (≈35° slope). The site names north and south reflect the location of the sites relative to the river, but not the aspect of the plots within the sites. The edaphic units were classified according to IUSS Working Group (2006); in the north site, the soil corresponds to a Vetic Acrisol (Humic, Clayic). The horizon sequence is of type Ap, Bt, BtC, C1, C2, of clay in texture. The depth of the solum is 70 cm while that of the organic horizon is 29 cm. In the south site, however, the soil is classified as a Haplic Acrisol (Humic), the horizon sequence is of type A1, BtC, C1, CR, of silty clay in texture. The depth of solum is 52 cm while that of the organic horizon is 10 cm. Both soil types contain remnants of volcanic ash that conserve some andic properties, while clay illuviation is present in the deep horizons. A coffee agroecosystem and a secondary forest plot were selected in the north and in the south sites (Fig. 1). Previous work in the same plots (Marín-Castro et al., 2016) showed that the soil bulk density (ρ_b) is ≤0.9 g/cm^3 (classified as low according to Shoji et al., 1993), while the antecedent soil moisture (θ_i, measured with the gravimetric method in the rainy season) has been recorded at above 40% v/v, clay content (extracted with the pipette method) ranges from 45 to 55% and the organic matter content has a range of 8 to 11% in the secondary forests and 7 to 16% in the coffee agroecosystems.

2.2. Description of the vegetation

The secondary forests considered in this study are fragments of TMCF of 2 to 3 ha, with a variable land use history (Fig. 1b). The secondary forest plot in the north (SFn) is derived from a pasture abandoned 30 years ago, the rapid regeneration of which has been driven by its owners through incorporation of plant species representative of the TMCF. The secondary forest plot in the south (SFs) was a shade coffee plantation abandoned 30 years ago that has undergone natural regeneration influenced by the surrounding forests. In the secondary forests, the arboreal stratum reaches an approximate height of 8–30 m and contains the genera *Quercus*, *Liquidambar*, *Macadamia*, *Trema*, *Inga*, *Heliocarpus*, *Clethra*, *Alchornea*, *Saurauia* and *Brunellia*. The shrub stratum has a height of 0.6–4.4 m and features the genera *Myrsine*, *Hoffmannia*, *Chiococca* and a species of *Cyatheaceae*. The herbaceous stratum can reach a height of 0.2–1.0 m and is represented by the family *Poaceae* and the genera of *Selaginella*, *Chamaedorea* and *Dichanthelium* (Castillo-Campos, 2012, unpublished data).

The land use history and management type of the two studied coffee agroecosystems also differ (Fig. 1b). The coffee agroecosystem of the north site (CAn) is derived from a pasture and has been under organic shade coffee management for the last 30 years, for which reason no herbicides have been applied and the weeds are controlled by manual

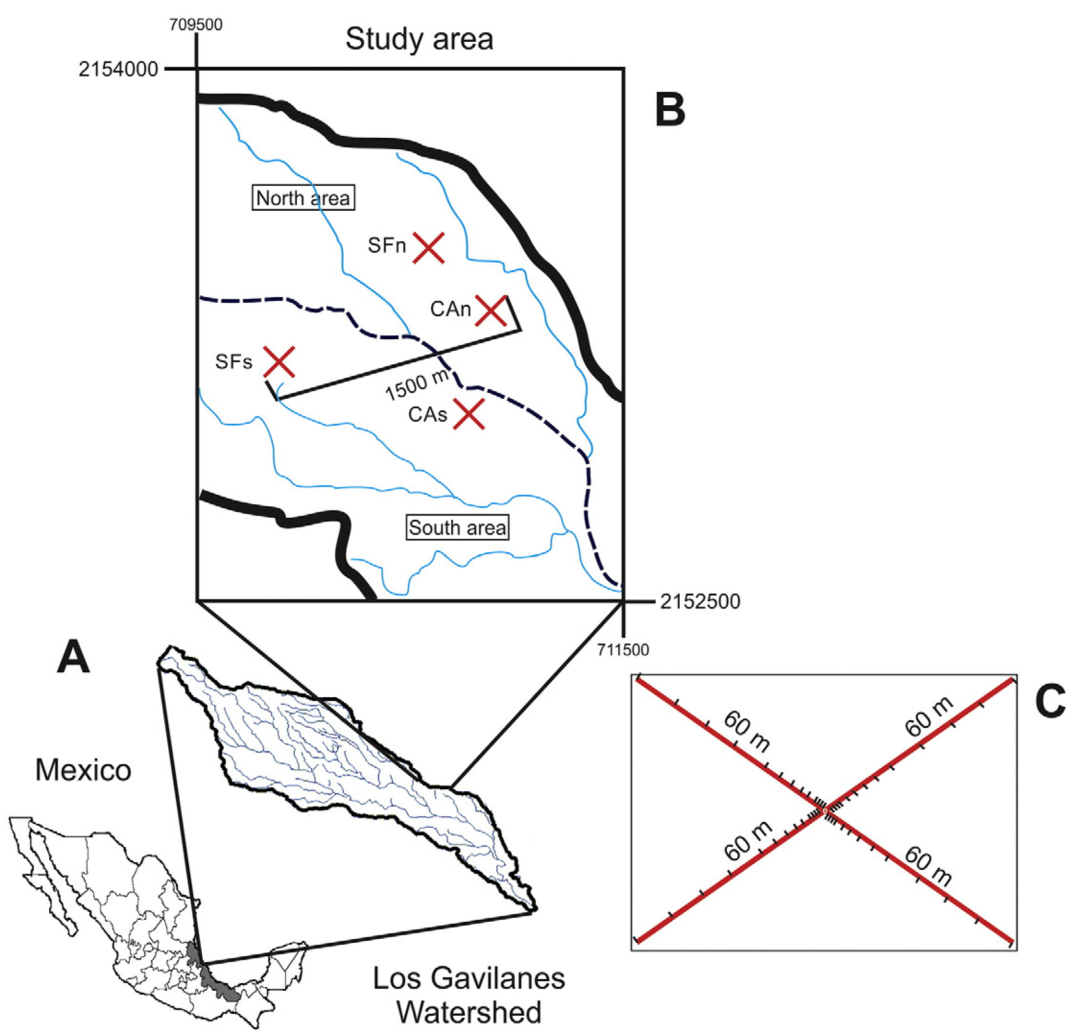

Fig. 1. Location of (A) Los Gavilanes watershed within Central Veracruz, Mexico; (B) study area and sampling plots within Los Gavilanes watershed; and (C) marks along the amplified X, indicating sampling locations. A red X indicates a sampling transects in either secondary montane cloud forest (SFn and SFs) or coffee agroecosystem (CAn and CAs), the 1500 m distance scale indicates the maximum distance between plots and the dashed line the Gavilanes River.

cutting. The coffee agroecosystem to the south (CAs) is derived from a natural forest and has had >80 years of intensive use and application of fertilizers with high phosphorous contents (according to the owner). The vegetal structure of both coffee agroecosystems consists mainly of two strata: the trees and the shrubs. The tree stratum ranges from 10 to 20 m in height and is represented by the genera *Inga*, *Liquidambar*, *Cecropia*, *Alchornea* and *Heliocarpus*, which produce 60% shade below the canopy. The shrub stratum is dominated by coffee plants (*Coffea arabica)* of 0.5–3 m in height and 220 (±24) plants per hectare in density (Williams-Linera and López-Gómez, 2008). Also found in the shrub stratum are ferns of the genus *Pteridium* and some species of the genus *Perrottetia* (Castillo-Campos, 2012, unpublished data).

2.3. Sampling design

Three criteria were used to select the plots: 1) the plots were situated on the same geoform (low hills) with slopes >35°; 2) they were located along the same altitudinal floor (1200–1300 m. a.s.l.) and 3) the land uses were secondary forest and shade coffee plantation. This study follows the definition of scale used by Dungan et al. (2002) and Negrete-Yankelevich and Fox (2015), comprising three magnitudes: interval (the minimum distance between observations), grain (the diameter of the sampling unit) and extent (the maximum distance between observations). The spatial distribution of K_{fs} was studied at two scales, differing only in the extent: 1) plot (within the secondary forest and coffee agroecosystems; with interval 0.5 m, grain diameter of 0.15 m (diameter of the infiltrometer ring) and extent of 120 m) and 2) landscape (interval of 0.5 m, grain diameter of 0.15 m and extent of 1500 m, the maximum distance between all observations in the study). In each of the plots, two crossed transects of 120 m each were established (Fig. 1b). The aperture angle of the cross was 40°, except for SF2 where it was 70°. This difference was determined by the shorter slope and the need to maintain a margin of 5 m of separation from other land uses. From the center of the cross, the distances 0, 0.5, 1, 2, 3, 5, 7, 10, 13, 16, 25, 37, 49 and 60 m were marked on each arm. This produced a total of 53 sampling points per plot (Fig. 1c). This design generated paired observations separated by a gradation of distances that (known also as lags), through a variographic analysis, enabled modeling the spatial structures of the variables of interest (Negrete-Yankelevich and Fox, 2015).

2.4. Measurement of field saturated hydraulic conductivity

From February to June 2012, in situ infiltration assays were conducted using a single ring infiltrometer (modified from Gómez-Tagle et al., 2008). Given that litter development along the transect length is heterogeneous, the infiltration assays were conducted directly on the first 5 cm in depth of the mineral soil. Data were collected automatically using a datalogger with differential pressure sensors, programed with free software and hardware of the platform Arduino (Banzi et al., 2010). The K_{fs} values were calculated using the flow values of the stable phase of the accumulated infiltration curve, following the Wu2 method (Wu et al., 1999) and using a value of 0.12 cm^{-1} for the alpha parameter, as suggested by Elrick and Reynolds (1992) for soils structured with clay and silts. Values of K_{fs} were expressed in mm h^{-1} (also see Xu et al., 2012). For more detail of these infiltration assays, see Marín-Castro et al. (2016). In some cases the infiltration tests occurred when the soil was saturated due to rain events the previous day. However, this was not considered influential because K_{fs} values were calculated with the flow values of the stable phase of the accumulated infiltration curve, when the soil reached a saturated phase in all cases. The influence of water repellency was also unlikely because soils in these TMCF (and coffee agroecosystems) have elevated soil water contents (Dekker, 1998; Marín-Castro et al., 2016).

2.5. Measurement of the litter layer and the root biomass

Measurement of the litter layer and soil sampling for root collection was conducted at the same time as the infiltration assays. Thickness of the litter layer was measured in a 0.5 m^2 quadrant using a ruler, considering this layer to comprise all material from the compact leaf litter down to the mineral soil layer detected with a hand-texturing test (Siebe et al., 1996). In this study, we used the layer thickness as an indicator of litter layer development. While this is often a less precise measurement than biomass, it is much easier to measure under difficult field conditions (Vorobeichik, 1997, 1995). In order to determine root biomass, the total volume of soil contained within the ring of the infiltrometer (1094 cm^3) was extracted, according to the method of Schuurman and Goedewaagen (1971).

Soil samples were stored in polythene bags and frozen at −20 °C until separation of the soil from the roots, in order to avoid degradation of root tissue. Separation of the roots began with defrosting of the samples at ambient temperature, then washing of the sample by submerging in water in order to separate the soil particles from the roots. The sample passed through three sieves of mesh sizes 2.4, 1.2 and 0.6 mm and the medium and thick roots were manually collected from each of these sieves with pincers. For extraction of the fine roots, the soil samples of the previous step were dried at temperature of 60 °C for 3 h and then spread out on a tray with a polythene sheet suspended 1 cm above the sample. The fine roots attached themselves to the polythene sheet through electrostatic attraction and were then collected. Using a vernier, the roots were classified according to diameter into three categories (following Campos et al. 2016): thick >2 mm, medium 1 to 2 mm and fine/very fine <1 mm. Once separated, the roots were weighed in an OHAUSS analytical balance with an error of ±0.0001 g.

2.6. Modeling of the spatial structures

In order to detect the spatial structures (gradients and patches) present in the explanatory variables (litter layer and root biomass), and K_{fs}, experimental variograms were constructed with each variable. Variograms are graphs of semivariance as a function of separation distance between all possible pairs of observations. Semivariance decreases as spatial correlation increases and is defined as:

$$\hat{\gamma}(h) = \frac{1}{2W(h)} \sum_{i=1}^{W(h)} \left[y_i - y_{i+h}\right]^2$$

where $\hat{\gamma}(h)$ is the semivariance of all of the pairs of observations located at any distance h, $W(h)$ is the total number of pairs of observations separated by h and y_i is the value of the sample in location i. For variograms to be reliable it is necessary to construct them using no >2/3 of the maximum distance available from the data (Negrete-Yankelevich and Fox, 2015).The distances between observations were therefore 0.5 to 80 m at the within plot scale and 0.5 to 1000 m at the landscape scale (which considers all 212 sampling points, Fig. 1b and c). The experimental variograms were modeled with theoretical linear, spherical, exponential and Gaussian variograms, the explanatory power of which was compared with the null (nugget) model (Fortin and Dale, 2005; Negrete-Yankelevich and Fox, 2015). For selection of the theoretical model, the change in the Akaike index (ΔAIC) between the fitted models and the nugget model was used as a criterion. When the fitted models presented a reduction smaller than five units relative to the null model, the null model was chosen (Richards, 2015). In order to avoid considering theoretical models in which the spatial pattern represents a very small portion of the total variation, only those variograms in which the C_I (modeled autocorrelation) was >30% of the total semivariance ($C_1 + Co$) were considered. In order to comply with the assumptions of the variogram, the variables were transformed to the best normalizing power, log or arcsine transformation (Tables 2 and 3). The interval estimated for variograms that reach asymptote (or the distance at which 90% of the asymptote is reached, if the model is asymptotic) is interpreted as the average diameter of the patches in which the variable is structured at the studied scale. When gradients (trends) were identified representing the gradual variation in space (Oliver and Webster, 2014), these were first modeled with a linear model using the north-south and east-west coordinates as explanatory variables and, subsequently, the residuals were modeled with theoretical variograms in order to model the patterns of patches within this gradient (Negrete-Yankelevich and Fox, 2015). The GeoR package was used to construct and model the variograms in R, v. 2.15.3 (R Core Team, 2014).

2.7. Linear modeling

In order to determine whether the thickness of the litter layer and the values of root biomass differed between land uses and area, a linear model was used, followed by treatment contrasts (between plots) with penalization of the alpha, through the Bonferroni method, in order to compensate for the increase in type 1 error when multiple comparisons are made. A linear model was also constructed in order to determine the explanatory power over K_{fs} of the thickness of the litter layer, the biomass of roots of the three root diameter categories, the land use type (secondary forest and coffee agroecosystem) and the location of the geographic area of the plots (north or south). We followed the procedures described by Negrete-Yankelevich and Fox (2015) for simultaneous model simplification and spatial structure detection and modeling in the residuals. First, a general model was fitted with all of the explanatory variables and then model simplification was conducted through progressive elimination of the non-significant ($p < 0.05$) variables (Chatterjee et al., 2000). In each one of the steps of model simplification, an analysis of variance (ANOVA) was performed in order to compare the explanatory power of the nested models (with and without each one of the variables). The change in Akaike index (ΔAIC) was used to estimate the parsimony of the models that explained the relationship of the variables with K_{fs}. In order to determine whether: 1) the simplified model complies with the assumption of spatial independence of the errors and 2) the independent variables could explain the spatial patterns of K_{fs}, variograms were constructed (and fitted with theoretical variograms) with the residuals of the different models. This was conducted first with the simplified model, and then with those models that included and excluded the variables that remained within the simplified model. If the residuals did not present spatial patterns when the model included the variable, but did present them on exclusion of the variable, it was considered that the variables included in the model explained both the average and the spatial structures (Negrete-Yankelevich and Fox, 2015). Finally, in order to determine whether the physical and biotic variables known to be explanatory of the K_{fs} are capable of making the factors of use and area redundant, we incorporated physical variables (clay and antecedent soil moisture content), found to be explanatory of K_{fs} in a previous study (Marín-Castro et al., 2016) into our simplified model, and compared this with a partial analysis of variance with similar models that excluded these factors of use and/or area. In order to identify whether the physical and biotic variables become redundant in their explanatory power of K_{fs} on co-variation, a final simplification of the model was conducted that included area, use and the physical and biotic variables, following the same procedure described above. All analyses were performed in the language and environment R, v. 2.15.3 (R Core Team, 2014).

3. Results

The litter in the four studied plots occurred in a thin layer and had diffuse borders with the underlying horizon, therefore it was classified as a mull type according to Siebe et al. (1996). The thickness of the litter layer ranged from 0 to 14 cm in the forests and from 0 to 13 cm in the coffee agroecosystems. The total root biomass in the forest ranged from 0.002 to 0.02 g cm^{-3} and in the coffee agroecosystems from 0.0015 to 0.024 g cm^{-3} (Table 1). In the models of the litter layer, total root biomass and thick root biomass, the factor use was significant, due mainly to the fact that SFn presented substantially higher values for these variables than was the case in the other plots. In particular, the average thickness of the litter layer was ca. 3 cm greater (double) in the SFn than in the rest of the plots. The biomass of thick roots was greater in SFn and the CAs than in the rest of the plots (Table 1). In the four plots, the hierarchical order of root biomass was fine roots > thick roots > medium roots. Abundance of medium roots was higher in the north site. We found no differences between land uses or area in terms of the biomass of the fine roots (Table 1). Unlike the K_{fs}, the litter layer presented a spatial structuring at plot scale in the south site (SFs and CAs), with an average patch diameter of 12 m in both plots (Table 2, Fig. 2).

Only the biomass values of the thick and fine roots of plot SFn presented a detectable spatial structure at plot scale, but this fitted a linear type theoretical variogram, indicating a patch size >80 m (Fig. 3). It was not possible to construct the variogram for the category of medium roots within the CAs plot because of the presentation of zero values in many of the sampling points. All of the studied variables presented significant gradients at landscape scale (Table 3, Fig. 4), but only the biomass of thick roots also presented patches of 585 m in diameter within the gradient. These patches concur with the linear autocorrelation signal detected within SFn (Table 3, Fig. 2). Values of K_{fs} showed a random distribution at the plot scale, but presented a gradient that increased from south to north at the landscape scale (Fig. 4).

Our simplified linear model of K_{fs} indicates that the variation of K_{fs} is explained by area (north-south), land use (secondary forest and coffee agroecosystem) and litter layer thickness (Table 1, Fig. 3), but not by any of the root biomass categories. On comparison between the explanatory power of the model that includes area, use and the physical and biotic properties of the soil, and those that exclude area and/or use, we found that both use and area continue to present a significant explanatory power ($F_{(1,209)} = 21.43$; $p < 0.0001$).

On simplifying the model that included the physical and biotic variables together with use and area, antecedent moisture was considered redundant due to elimination by the greater explanatory power of the litter layer. The residuals of the models produced by this simplification, and of those that explain the K_{fs} with each variable individually, did not present autocorrelation in space (variograms with the nugget model). For this reason, the spatial structure of a gradient presented by K_{fs} at landscape scale can be explained by the influence of area, use, litter layer thickness or percentage of clay in the soil (Table 3, Fig. 4).

4. Discussion

4.1. Differences in the average thickness of the litter layer and root biomass

The litter layer thickness values obtained here for the forests and coffee agroecosystems are similar to those reported in the montane forests of Malaysia (3–8 cm) (Jeyanny et al., 2014) and are within the 5 to 20 cm interval reported for other TMCF of the region (e.g., Campos et al., 2016). Our study found no differences between land uses, i.e., the input of organic matter in both ecosystems was similar among the plots studied. The greater development of the litter layer presented in SFn compared to the other plots may be due to the fact that the owners in this case had been actively introducing native deciduous or sub-deciduous tree-shrub species for the last 30 years, which was not the case in SFs. Other studies have documented that in coffee agroecosystems (Sharma et al., 1997), cocoa crops and cultivated oak (Pandey et al., 2007), both the form and intensity of management cause a variation in the quantity and quality of the litter (Kumar, 2007). Indeed, pruning of the coffee plants, or even defoliation as a result of the fungal disease Roya (*Hemileia vastatrix*), mean that much more litter and high quality organic material is concentrated in the

Table 1
Results of linear models that explain the saturated hydraulic conductivity (K_{fs}) and the vegetation variables that influence this parameter. The table presents the mean values and coefficient of variation in the thickness of the litter layer and root biomass in two sites of secondary forest (SFn and SFs) and two sites of shade coffee (CAn and CAs) in the sub-basin of the Los Gavilanes River in Veracruz, Mexico, during the period February to June 2012.

Explanatory variables in the final model						North				South			
Response variable	Area	Use	Litter layer	Clay	ΔAIC	SFn		CAn		SFs		CAs	
	$F_{(1,208)}$	$F_{(1,208)}$	$F_{(1,208)}$	$F_{(1,208)}$		Mean	CV	Mean	CV	Mean	CV	Mean	CV
$K_{fs}^{\ddagger}$	8.27**	34.60***	42.79***	25.06***	0.47	1245.8^{a}	47	570.7^{b}	105	649.8^{b}	104.6	294.4^{c}	89.7

Explanatory variables in the final model					North				South			
Response variable	Area	Use	Litter layer	ΔAIC	SFn		CAn		SFs		CAs	
	$F_{(1,209)}$	$F_{(1,209)}$			Mean	CV	Mean	CV	Mean	CV	Mean	CV
Litter layer	24.7***	6.7*	na	104.8	6.3^{a}	47	3.3^{b}	100	2.1^{b}	96	3.0^{b}	64
TM	0.002	5.54*	na	−2.0	15.7^{a}	18	7.2^{b}	54	8.6^{b}	48	13.8^{b}	23
Rt	0.07	8.8**	na	−1.93	5.4^{a}	34	2.0^{b}	82	3.2^{b}	87	4.5^{a}	50
Rm	0.5	0.001	na	–	3.4^{a}	22	1.1^{b}	70	0.8^{b}	71	3.1^{a}	21
Rf	0.007	3.21	na	–	6.9^{a}	28	4.1^{b}	79	4.6^{b}	55	6.1^{a}	25

$K_{fs}^{\ddagger}$ = field saturated hydraulic conductivity (mm h^{-1}); data taken from Marín-Castro et al., 2016, modified for comparative purposes. Rt = thick roots (>2 mm) (kg m^{-3}); Rm = medium roots (1–2 mm) (kg m^{-3}); Rf = fine roots (<1 mm) (kg m^{-3}); TM = total biomass (g cm^{-3}); CV = coefficient of variation in percentage. ΔAIC = change in the Aikake index. –No variable was found with explanatory power and there was therefore no ΔAIC. Different letters denote significant differences under the test of contrasts, followed by a Bonferroni correction. na = not applicable.
* $p < 0.01$.
** $p < 0.001$.
*** $p < 0.0001$.

soil in the coffee plantations (FAO, 2015), at least for limited periods at the time. These inputs in the long-run may cause an increase in soil organic matter content and alter its decomposition and quality, ultimately hampering water conduction in the soil (through repellency) when water content is low (Goebel et al., 2011). In our case (with high water content), the fact that the litter layer was a variable with explanatory power over K_{fs} and that both variables presented their maximum values in SFn, suggests that active restoration could contribute to promoting water conductivity. Likewise, it has been found that management differences have consequences for some ecosystem functions, such as carbon capture (Youkhana and Idol, 2009) or biodiversity of the soil fauna (Perfecto et al., 1996).

For root biomass, we only found a difference among plots for the thick roots (>2 mm), which had greater biomass in the SFn and in the CAs. In contrast, in Indonesian forests, anthropic disturbance can reduce the biomass of fine roots (Leuschner et al., 2006). Our study only considered the first five centimeters of the soil, but it is known that the root biomass of species of forest and coffee plantations develops in the first 30 cm of soil (Defrenet et al., 2016). It is therefore possible that there are differences at greater depths that cannot be observed at the surface. However, our finding regarding the thick roots is consistent with that proposed by Defrenet et al. (2016), in which some of the tree species that provide shade to the coffee should reduce the coffee plant root biomass through competition. It is also possible that, in some forests, the

Table 2
Parameters estimated for the variograms of the litter layer thickness and the biomass of thick (Rt), medium (Rm) and fine (Rf) roots and total root biomass (TB) in two sites of secondary forest (SFn and SFs) and two sites of shade coffee (CAn and CAs) in the sub-basin of the "Los Gavilanes" River in Veracruz, Mexico.

Site/variable	Transformation	Model	Nugget	Interval	Least squares	r^2	Co + C
SFn							
Litter	$\sqrt{[\text{arcsine}(Y / 100) - 0.5]}$	Nugget	0.00	>80	86.64	0.90	8.55
Rt	log(Y − 1)	Linear	2.77	>80	1086	0.46	2.81
Rm	$(Y)^4$	Nugget	0.66	>80	2.58	0.90	0.66
Rf	log(Y)	Linear	0.00	>80	7779	0.83	3306
TB	$\sqrt{Y}$	Nugget	0.14	>80	0.03	0.99	0.14
SFs							
Litter	$\sqrt{Y}$	Spherical	0.01	**11.9**	11.54	0.60	0.48
Rt	$\sqrt{Y}$	Nugget	0.64	>80	0.38	0.98	0.64
Rm	log(Y)	Nugget	0.52	>80	0.51	0.97	0.52
Rf	log(Y)	Nugget	0.30	>80	0.09	0.99	0.30
TB	$\sqrt{Y}$	Nugget	0.53	>80	0.25	0.99	0.53
CAn							
Litter	$\sqrt{[(Y / 100)]}$	Nugget	0.01	>80	0.00	0.98	0.01
Rt	$\sqrt{Y}$	Nugget	0.34	>80	0.11	0.97	0.34
Rm	$(Y)^2$	Nugget	2.97	>80	6.64	0.97	2.97
Rf	$\sqrt{Y}$	Nugget	0.01	>80	0.00	0.98	0.01
TB	$\sqrt{Y}$	Nugget	0.41	>80	0.98	0.97	0.41
CAs							
Litter	$\sqrt{Y}$	Exponential	0.03	**12.4**	6.07	0.40	0.34
Rt	$\sqrt{(Y)^{-4}}$	Nugget	0.53	>80	1.05	0.93	0.53
Rm	–	–					
Rf	$\sqrt{Y}$	Nugget	0.00	>80	0.00	0.98	0.00
TB	na	Nugget	11.97	>80	414.35	0.97	11.97

Bold numbers correspond to the variogram intervals (mean patch size) for variables that show a significant spatial autocorrelaton within plots.
na = not applicable, distribution is normal.

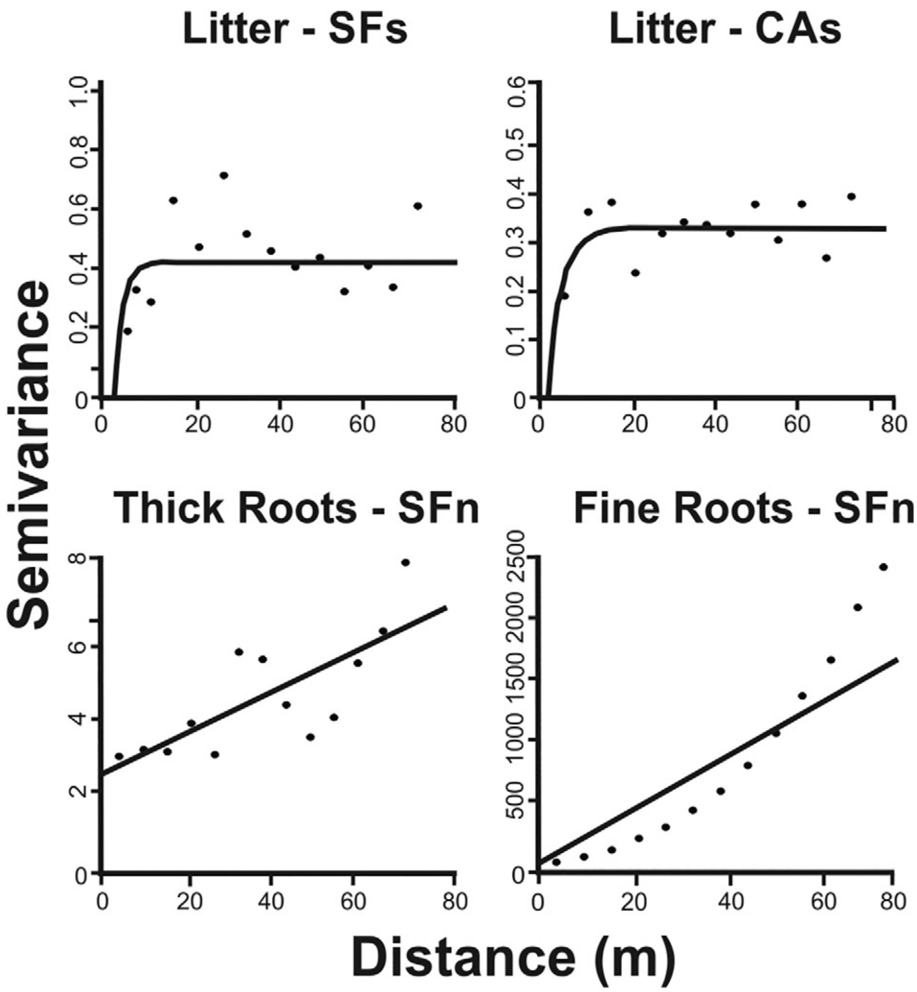

Fig. 2. Semivariograms of the variables litter layer and roots. Only those that fitted to a theoretical model are shown. Table 2 shows the parameters of all of the variograms at plot scale.

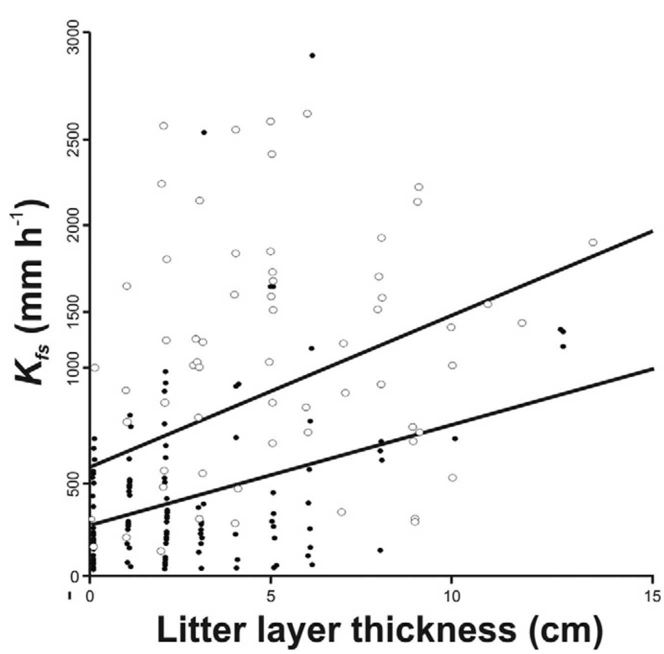

Fig. 3. Relationship between the thickness of the litter layer and the K_{fs} in secondary forests (white circles) and shade coffee plantations (black circles) in the sub-basin of the Los Gavilanes River, in Veracruz, Mexico. The lines correspond to the predictions of the final linear model.

vegetation develops thicker roots that can favor mechanisms of anchorage for stabilization, particularly on steep slopes (Soethe et al., 2006).

A greater abundance of fine roots was invariably found in the four plots, followed by thick roots and finally medium roots. The values of biomass of fine roots found are similar to those reported in the TMCF of the region (2.86–11.59 kg m^{-3}) (Campos et al., 2016) and in Indonesia (2.5–5 kg m^{-3}; Leuschner et al., 2006). It is probable that the greater development of fine roots in the surface of these ecosystems is due to the fact that the plants have invested their resources in the acquisition of nutrients that are available ephemerally in the soil surface layer through decomposition of leaf litter (Padovan et al., 2015). As in our case, in Costa Rica, transformation of the TMCF to a low impact system of cultivation such as shade coffee agroecosystems, acts to conserve part of the vegetable coverage of the original forest as well as the hydrophysical (moisture, high porosity and organic matter) and biotic (a high proportion of the plant biomass in the soil) properties (Defrenet et al., 2016). Nevertheless, the fact that greater differences were found between plots and not between uses supports the findings of other studies, in which alteration of the soil properties depends substantially on management type (intensive or semi-intensive) (Hernández-Martínez et al., 2009; Moguel and Toledo, 1999).

4.2. Relationship between K_{fs} and the litter layer and roots

The simplified model of biotic variables that explain K_{fs} included study area (north and south), land use type (forest and coffee plantation) and thickness of the litter layer. Even including the physical variables (antecedent moisture and percentage of clays), we could not

Table 3
Parameters of the gradient models and the variograms that describe the spatial structures of the thickness of the litter layer and biomass of thick (Rt), medium (Rm) and fine (Rf) roots at landscape scale.

Variable	Model of gradient			Models of the variogram					C1/C1 + Co
	a	*b*	ΔAIC	Model	Range	ΔAIC	Nugget (Co)	C1	
Litter layer	-8.244×10^{-4}***	-5.742×10^{-4}**	481.3	Nugget	0	4.2	0	0.6841	–
Rt	-1.78×10^{-5}***	-2.50×10^{-5}***	−1029.8	Exponential	585	44.1	0.0004	0.0001	–
Rm	-1.30×10^{-5}***	-4.27×10^{-5}***	−1110.8	Nugget	0	–	0	0.0004	–
Rf	-3.46×10^{-4}***	-6.99×10^{-4}***	271.4	Nugget	0	–	0	0.2348	
Parameters of the variograms constructed with the residuals of the linear model									
$K_{fs} = \alpha$ Litter				Nugget	0	–	4.051	0	0
$K_{fs} =$ x use				Nugget	0	–	1.017	0	0
kfst = α Litter + β clay + x area + y use)				Nugget	0	–	3.228	0	0

The parameters *a* and *b* denote the coordinates north–south and east–west, respectively. kfst = $(K_{fs})^{0.3}$.
** p< 0.01.
*** p< 0.001.

eliminate the area and use from the model. This indicates that there are differences between north and south and between secondary forests and coffee agroecosystems that influence K_{fs} and which to date have not been studied. The differences between the north and south areas are probably related to pedogenetic processes, considering that this is a young and dynamic edaphic system (Dubroeucq et al., 2002). There may also exist differences between north and south in terms of root biomass at greater depths and the root structure of the dominant plant species (Allaire et al., 2009). Unexplained differences between land uses may be related to lower macroporosity and pore continuity in coffee agroecosystems as a result of human traffic that compacts the soil (Alaoui and Goetz, 2008; Meek et al., 1992). The lack of replicates of land use within the two areas prevented the detection of possible differences between north and south in terms of the impact of land use (area-use interaction) and, for this reason, we can only hypothesize about the origin of the differences found among plots. Our results appear to covary with a possible gradient of land use intensity. The SFn, with the highest K_{fs} (and litter layer development) found in the study, correspond to a secondary forest that has been restored and enriched with native species. The plots of CAn and SFs, with intermediate K_{fs} values correspond to a forest derived from an abandoned coffee plantation (SFs), which was left to natural succession for approximately 30 years, and CAn with a non-intensified use, since it is almost exclusively for autoconsumption and presents little management intervention on the part of the owners and where natural succession has not been completely eliminated. Finally, the CAs with the lowest value of K_{fs} corresponds to a plot with commercial coffee plantations of over 80 years in age and intensive use (two manual harvests are produced per year). It is notable that the litter layer, while explaining on average the K_{fs}, does not follow the same average pattern among plots. Its explanatory power over K_{fs} could be associated with variation within the plots, where it is very heterogeneous (CV = 47–100%). This variation may be due to differences in leaf litter production among tree species. Williams-Linera and Tolome (1996) indicate that trees of the TMCF differ substantially in production depending on phytogeographic origin: Holarctic species contribute a greater volume of leaf litter, which

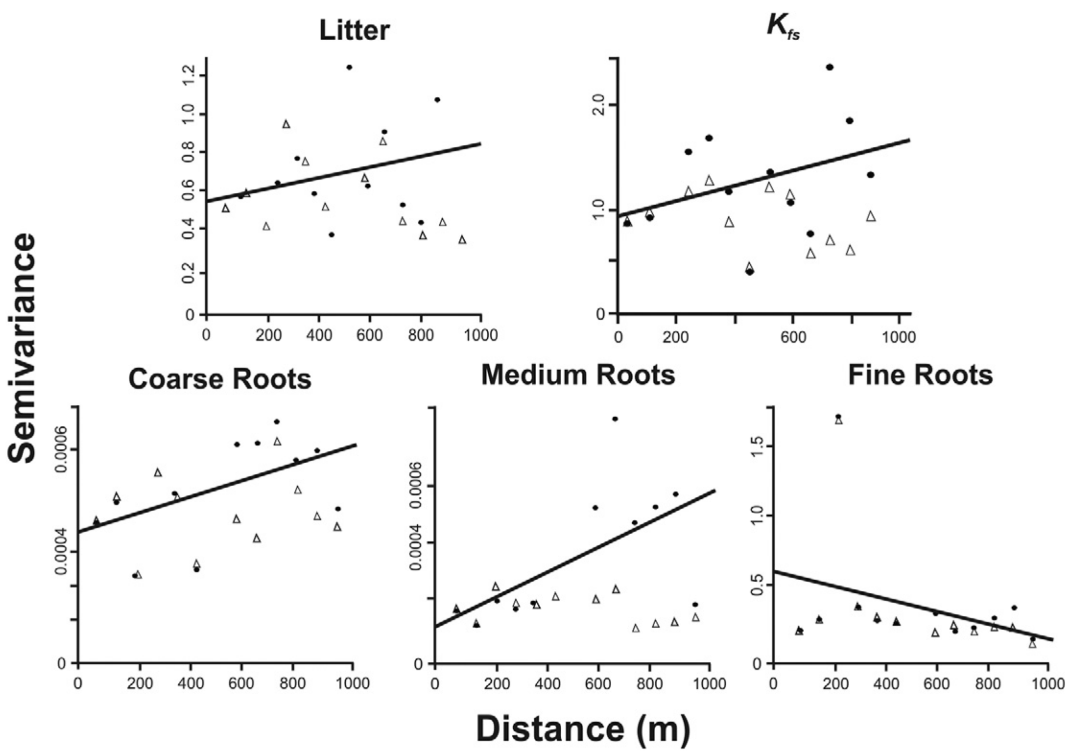

Fig. 4. Semivariograms at landscape scale of the K_{fs}, litter layer and different categories of roots that explain the K_{fs}. The black circles represent the semivariance including gradients. The triangles are semivariants, excluding the gradients (detrended). The solid lines are the fitted linear models that represent the gradient. The dotted lines are the fitted theoretical variographic models where the asymptotic behavior of the semivariance is detected.

could be reflected in the patched pattern of litter within the plots. The positive relationship between litter layer thickness and K_{fs} suggests that the litter layer could maintain a surface and local microclimate (humidity and temperature) that indirectly keeps the soil pores active, facilitating water conduction as has been demonstrated by experimental studies in the field and laboratory (Basile et al., 2003). The fact that the litter layer may make antecedent moisture redundant in the model of K_{fs} supports this notion and also suggests that the effect of the litter layer on the K_{fs} could be reflected not in the instantaneous soil water content (as reflected by antecedent moisture) but rather in a more constant humidification over time (Boyd and Van Acker, 2003).While we did find differences among plots in terms of the biomass of thick roots, none of the characteristics associated with the roots are included in the model that explains K_{fs}. If, according to that proposed by other studies, the roots play an important role in the formation of channels of preferential water flow (Lin and Zhou, 2008; Simunek et al., 2003), it is probable that these occur at greater depths over a longer time frame (Allaire et al., 2009).

4.3. Spatial structure of the litter layer thickness and root biomass

We hypothesized that the spatial structuring of the litter layer and the root biomass would occur at landscape scale and that these variables would statistically explain the spatial variation of K_{fs}. While we found that, on average, root biomass did not explain K_{fs}, at landscape scale, the thickness of the litter layer and the biomass of the thick, medium and fine roots presented a spatial distribution gradient that was similar to that of the K_{fs}. The thick roots were also structured in patches of 585 m within this gradient. At plot scale, the litter layer thickness was structured in patches of twelve meters, but in the plots of SFs and CAs only. This spatial variation of the litter layer at plot scale has been explained by the heterogeneity of the composition of the forests and of the shade trees in the coffee plantations, as well as by the microtopography of the soil (Dunne et al., 1991; Facelli and Pickett, 1991). It is possible that spatial patterns at this scale may be affected by the specific local influence in which certain plant species present a differential (in time and space) input of organic material (leaf litter) within the plot (Corti et al., 2002; Freschet et al., 2012; Negrete-Yankelevich et al., 2007; Titus and del Moral, 1998), but these differences do not occur in the same manner in the north area as they do in the south area, and they do not produce an equally patched distribution of K_{fs}.

5. Conclusion

The impact of anthropogenic transformation of ecosystems on the vegetation affects ecosystems services of great importance such as the capacity to capture and store water. Our best model indicates that geographic area, land use, clay content and thickness of the litter layer explained K_{fs} and its spatial distribution on a gradient. The greater explanatory power of the litter layer, compared to antecedent moisture, could be related to the creation of a microenvironment that maintains more permanently soil pores in a functional state. Our modeling also suggests that differences in K_{fs} between the coffee agroecosystems and secondary forests of central Veracruz are multifactorial (biotic and physicochemical) and that differences prevail between the uses and geographic areas (possibly related to edaphogenesis, history of use and management practices) that determine the K_{fs}. These differences, however, remain to be fully studied.

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